

Qatar-4b

R-2519

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_211113 · created 2026-08-25T00:58:12+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459531.6064 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.027
Transit depth (Rp/Rs)^2	3.36 +/- 1.0 [%]
Orbital Inclination (inc)	86.31 +/- 1.78
Airmass coefficient 1 (a1)	1.334 +/- 0.035
Airmass coefficient 2 (a2)	-0.27 +/- 0.1
Scatter in the residuals of the lightcurve fit is	2.63 %
Best Comparison Star	#3 - [63, 92]
Optimal Method	PSF photometry
Transit Duration (day)	0.083 +/- 0.0074

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.183 ± 0.027 vs 0.138 (deviation 1.67σ)
Duration fit vs published	0.083 d vs 0.0892 d (deviation 0.83σ)
Mid-time O–C	6.3 min
Depth SNR	6.8
Residual scatter	2.63 %

Light curve

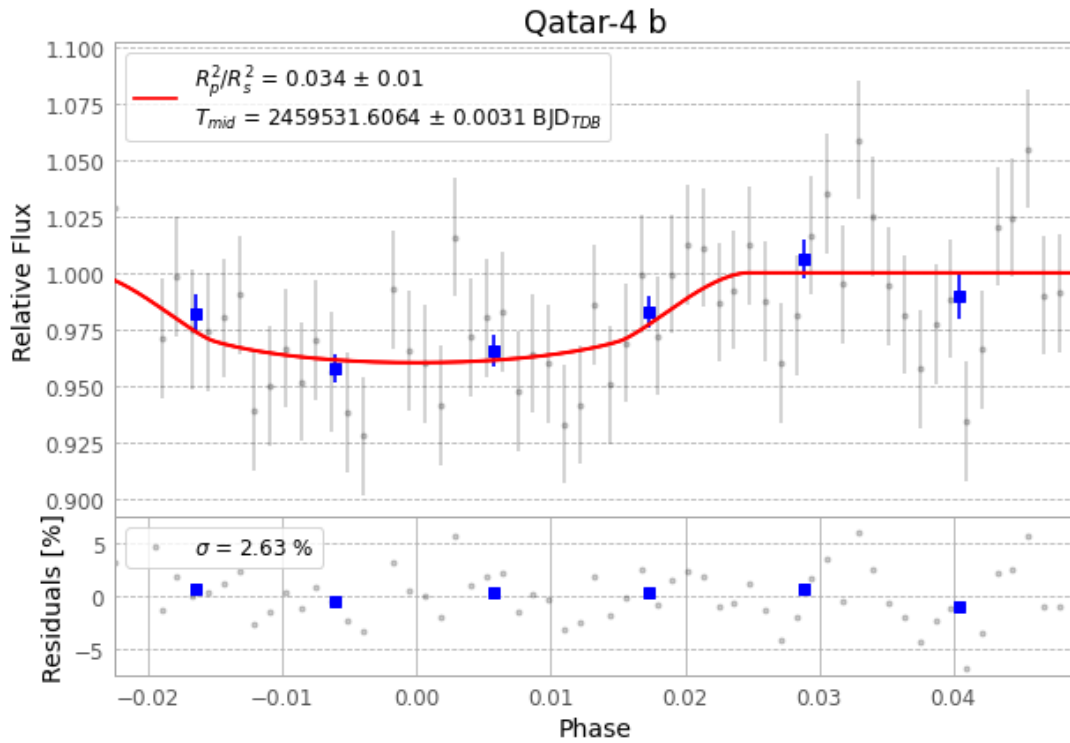


Figure 1 — FinalLightCurve_Qatar-4 b_2021-11-12.png (SHA-256 2cd0fe2e8222b0b3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 232], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e31971fc86733f08...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), Qatar-4211113012710.FITS → Qatar-4211113043312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-211113.log (79b3c4d8929c...), FinalLightCurve_Qatar-4 b_2021-11-12.png (2cd0fe2e8222...), FinalLightCurve_Qatar-4 b_2021-11-12.csv (abc0f2c8271c...)

Compute resources

Wall time 118.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 00:58Z.